

ROCHA, Uruguay

WMO: 865650

Lat: 34.4935S

Lon: 54.3126W

Elev: 63

StdP: 14.66

Time Zone: -3.00 (W03)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	33.8	36.2	30.1	24.2	39.5	32.1	26.6	40.5	23.3	56.4	19.9	55.2	0.9	350	0.386

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	19.1	88.8	72.3	86.0	71.6	83.4	70.9	75.6	84.0	74.2	81.9	72.9	79.9	8.1	0

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
73.2	123.5	79.4	71.9	118.0	77.9	70.7	113.5	76.7	39.2	84.2	37.8	81.9	36.7	80.0	82.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 20.3	17.7	15.3	29.5	96.1	1.4	3.5	28.5	98.6	27.6	100.6	26.8	102.6	25.8	105.1		
(5)			29.2	78.2	1.4	2.2	28.2	79.8	27.4	81.1	26.6	82.3	25.5	84.0		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	61.4	71.9	70.9	67.7	63.3	57.4	52.8	51.6	53.7	55.2	59.8	64.1	68.9		
	DBStd	9.14	5.24	5.10	5.22	5.73	5.76	6.12	6.71	6.69	6.05	5.58	5.65	6.06		
	HDD50	149	0	0	0	0	5	34	59	34	16	1	0	0		
	HDD65	2126	6	8	32	98	244	370	420	357	300	178	83	30		
	CDD50	4314	678	586	549	399	235	119	109	149	173	306	424	587		
	CDD65	814	219	174	116	47	9	4	5	7	7	18	56	152		
	CDH74	5613	1706	1148	652	176	30	11	20	49	54	122	419	1226		
	CDH80	1509	506	315	125	26	2	0	2	9	12	23	94	395		
(14) Wind	WSAvg	5.6	5.9	5.3	5.1	4.9	4.5	5.1	5.2	5.8	6.4	6.3	6.5	6.5		
(15) Precipitation	PrecAvg	43.5	3.9	4.0	4.3	4.0	3.8	4.0	4.3	3.7	3.8	3.9	3.5	2.8		
	PrecMax	63.3	11.1	11.2	17.6	19.2	11.7	12.9	9.5	14.3	8.6	10.2	9.1	10.6		
	PrecMin	25.8	0.2	0.2	0.5	0.4	0.7	0.1	0.6	0.7	1.0	0.7	1.0	0.1		
	PrecStd	10.2	2.5	2.7	3.3	3.5	2.2	2.7	2.3	2.3	2.0	2.3	2.1	2.0		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	92.7	91.9	87.3	84.0	77.7	75.0	77.0	80.0	80.2	83.0	87.3	92.9		
		MCWB	73.8	73.5	71.9	72.7	66.6	66.7	65.6	67.5	68.7	68.0	70.6	73.0		
	2%	DB	88.3	87.0	83.3	78.7	73.1	69.8	69.6	73.1	72.9	77.0	82.6	87.9		
		MCWB	72.9	73.1	70.6	69.3	64.3	63.7	62.6	63.3	62.8	65.9	68.8	70.5		
	5%	DB	85.2	83.7	80.4	75.4	69.7	65.8	64.9	68.0	68.8	73.2	78.8	84.3		
		MCWB	72.1	71.9	70.1	67.5	63.2	60.6	59.8	60.6	60.6	64.3	66.9	70.1		
	10%	DB	82.4	80.5	77.6	72.6	66.8	62.8	61.8	64.2	65.5	69.8	75.1	80.8		
		MCWB	71.3	70.5	69.0	66.5	62.1	58.4	57.7	59.0	58.9	62.3	65.4	69.0		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	78.2	77.3	75.7	73.6	68.6	68.3	66.5	68.8	70.4	70.7	73.7	76.5		
		MCDB	86.1	86.5	82.3	81.2	74.5	73.7	74.3	77.3	79.1	78.9	83.1	87.2		
	2%	WB	75.8	75.0	73.5	71.4	66.3	64.9	63.5	65.3	65.7	68.0	70.7	73.9		
		MCDB	84.0	83.3	79.5	76.8	71.0	68.8	68.4	71.1	71.2	74.5	79.0	83.1		
	5%	WB	74.2	73.5	71.9	69.7	64.7	62.2	61.1	62.5	62.6	65.7	68.8	72.1		
		MCDB	81.7	80.7	77.4	73.2	68.2	65.3	64.3	66.2	66.2	70.5	75.5	80.3		
	10%	WB	72.8	72.1	70.5	67.9	63.0	59.6	58.8	60.0	60.5	63.9	67.0	70.6		
		MCDB	79.4	78.2	75.7	71.1	66.1	62.0	61.3	63.2	64.4	68.4	73.0	77.7		
(35) Mean Daily Temperature Range	5% DB	MDBR	19.1	17.5	17.7	16.2	16.5	16.3	16.0	16.1	17.1	17.3	18.9	19.7		
		MCDBR	22.9	21.8	21.7	20.0	19.7	19.0	18.4	21.7	22.6	22.7	24.7	25.9		
		MCWBR	10.3	10.3	10.8	11.1	11.8	12.8	11.7	12.9	13.3	12.9	12.9	12.0		
	5% WB	MCDBR	17.6	17.7	16.9	14.3	15.8	14.8	14.5	17.7	16.6	17.7	19.7	20.5		
		MCWBR	9.2	9.5	9.7	8.9	10.3	11.1	10.4	11.8	11.5	11.5	11.7	10.6		
(40) Clear-Sky Solar Irradiance	taub		0.399	0.403	0.373	0.373	0.360	0.363	0.370	0.449	0.463	0.416	0.387	0.387		
	taud		2.386	2.394	2.466	2.412	2.425	2.399	2.341	2.071	2.053	2.251	2.366	2.396		
	Ebn at Noon		298	289	286	266	252	240	244	235	253	282	299	303		
	Edh at Noon		41	39	34	31	28	26	30	43	50	44	41	40		
(44) All-Sky Solar Radiation	RadAvg		2260	1933	1599	1181	831	703	732	962	1315	1710	2134	2304		
	RadStd		172	184	140	132	56	64	65	95	97	153	177	157		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (5 neighbors)	+0.86	+1.26	N/A	+1.24	+1.13	N/A	N/A	N/A	+225	+121

Nomenclature: See separate page